

WINSTON FIELD, TX, USA

WMO: 722122

Lat: 32.693N

Lon: 100.951W

Elev: 2430

StdP: 13.45

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 03050

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.4	22.7	-3.8	4.9	53.9	2.0	6.6	50.1	30.4	54.1	27.2	50.4	8.0	20	0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	100.1	67.9	97.7	68.5	95.5	68.8	75.0	89.4	73.9	87.9	73.0	86.6	11.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.8	128.9	79.4	70.4	122.5	78.3	69.6	119.0	77.9	40.2	89.6	39.2	88.4	38.2	86.2	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	22.7	19.7	12.4	103.3	4.4	3.6	9.2	105.8	6.6	107.9	4.1	109.9	0.9	112.5
			10.8	77.2	4.2	1.8	7.7	78.5	5.2	79.6	2.9	80.7	-0.2	82.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.8	43.7	47.8	56.7	64.2	71.9	80.1	82.1	81.7	74.7	64.1	53.1	44.3
	DBStd	16.08	9.27	10.77	9.46	8.85	8.55	5.19	4.33	4.71	6.22	8.71	8.97	9.40
	HDD50	748	234	154	45	9	1	0	0	0	0	11	70	224
	HDD65	2714	660	485	283	118	36	0	0	0	6	119	365	642
	CDD50	5781	40	91	254	436	679	904	995	982	741	449	163	47
	CDD65	2269	0	3	26	94	248	454	530	517	297	92	8	0
	CDH74	26131	17	106	408	1210	2755	5260	6437	6102	2826	869	122	19
	CDH80	13070	0	19	109	488	1334	2783	3485	3311	1241	282	18	0
Wind	WSAvg	9.7	9.6	10.3	11.3	12.0	10.7	10.2	8.5	8.1	8.0	9.0	9.0	9.2
Precipitation	PrecAvg	22.2	0.7	1.0	1.2	1.5	3.0	3.1	1.9	2.5	3.0	2.4	1.2	0.7
	PrecMax	42.8	2.4	4.5	6.1	5.9	8.9	7.8	6.3	16.7	12.9	6.7	8.1	4.4
	PrecMin	12.1	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	6.3	0.7	0.9	1.3	1.6	2.4	2.0	1.8	3.1	2.7	1.9	1.6	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.5	83.0	88.8	95.0	100.4	103.6	101.5	102.2	97.4	90.8	82.2	76.9
		MCWB	52.4	55.5	56.9	56.9	61.6	65.4	69.8	68.5	67.8	67.0	58.7	52.4
	2%	DB	70.7	77.1	82.7	90.3	96.0	99.6	99.1	99.7	94.3	87.7	77.3	70.2
		MCWB	49.6	52.7	55.6	58.5	63.3	68.1	69.1	69.4	68.3	65.7	57.7	51.8
	5%	DB	65.6	72.2	78.9	86.9	92.2	96.3	96.7	97.1	90.7	82.4	73.4	64.5
		MCWB	47.7	52.1	55.9	59.3	65.1	69.2	69.8	69.6	68.4	64.2	56.2	50.0
	10%	DB	59.8	66.8	74.5	81.1	88.7	93.8	94.8	94.8	88.1	79.0	69.1	59.7
		MCWB	46.4	50.2	55.1	58.9	65.3	69.6	70.1	69.9	68.3	62.6	54.7	48.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.3	61.3	66.2	69.1	73.5	76.8	76.5	76.3	75.0	72.3	64.5	60.3
		MCDB	63.2	71.2	74.2	80.0	85.8	91.3	90.7	89.9	87.4	82.5	73.4	65.4
	2%	WB	53.8	58.2	63.9	67.1	71.9	74.7	74.6	74.5	72.9	70.3	61.8	56.1
		MCDB	63.1	68.8	73.3	76.5	83.6	89.3	89.0	88.9	83.9	79.8	70.9	62.8
	5%	WB	50.5	54.9	61.4	65.3	70.7	73.4	73.4	73.3	71.6	68.6	59.4	52.2
		MCDB	63.2	66.3	70.3	75.0	81.9	87.3	87.6	87.9	82.2	77.5	67.5	61.9
	10%	WB	47.3	51.5	58.8	63.0	69.3	72.3	72.4	72.3	70.5	66.6	56.9	48.8
		MCDB	58.4	64.6	68.9	73.9	80.1	85.9	86.4	86.6	80.8	75.0	66.1	59.9
Mean Daily Temperature Range	5% DB	MDBR	24.8	25.4	25.8	26.2	23.2	22.7	21.7	21.8	20.8	23.0	24.3	22.7
		MCDBR	34.4	34.7	31.8	32.5	29.0	26.7	24.7	25.1	25.2	27.6	30.0	31.7
		MCWBR	17.4	17.0	14.2	13.6	12.1	9.2	8.1	7.9	8.4	12.4	14.4	17.0
	5% WB	MCDBR	29.5	28.7	24.8	24.1	23.7	23.3	21.4	21.9	19.7	22.3	23.3	25.9
		MCWBR	16.8	17.3	13.6	13.7	11.0	9.0	7.9	7.8	7.4	11.8	14.0	16.1
Clear-Sky Solar Irradiance	taub	0.276	0.287	0.315	0.344	0.357	0.394	0.399	0.388	0.360	0.337	0.304	0.284	
	taud	2.563	2.538	2.470	2.393	2.430	2.359	2.409	2.407	2.439	2.486	2.509	2.542	
	Ebn at Noon	304	310	306	299	294	282	280	283	287	287	290	294	
	Edh at Noon	25	29	33	38	37	39	37	37	34	30	27	25	
All-Sky Solar Radiation	RadAvg	1052	1288	1680	2063	2177	2339	2243	2073	1715	1384	1103	924	
	RadStd	108	153	119	139	195	142	137	111	121	180	122	91	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (15 neighbors)		+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+218	+149

Nomenclature: See separate page